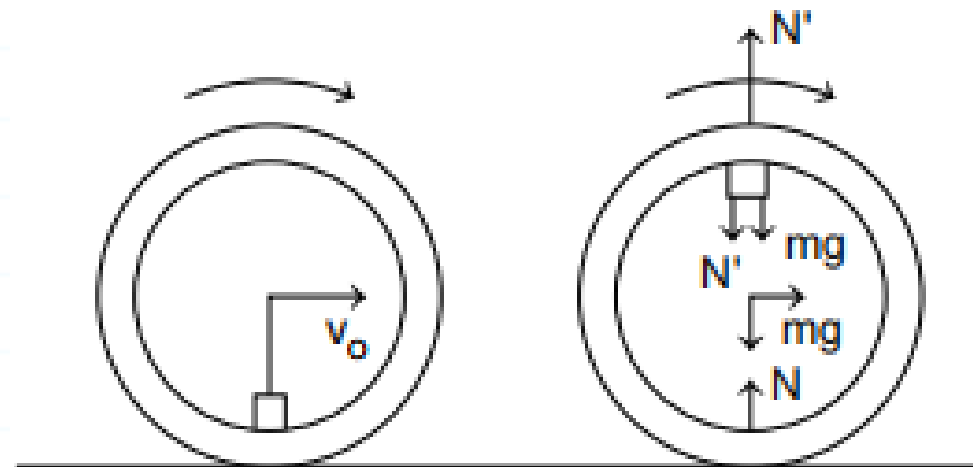


☞ **Solution** Clearly, the tendency of bouncing of the loop will be maximum when the small body A , will be at the highest point of the hoop during its rolling motion. Let the velocity of centre of mass of the hoop equals v at this position. The static friction does not work on the hoop, so from conservation of energy.



or $N = mg - N'$

$$\begin{aligned}
 &= mg + mg - \frac{mv^2}{R} = 2mg - \frac{mv^2}{R} \\
 \Rightarrow v^2 &= \frac{2mgR - NR}{m} \\
 &= 2gR - \frac{NR}{m} \quad \dots(iii)
 \end{aligned}$$

From equations (i) and (iii), we get

$$3 \left[2gR - \frac{NR}{m} \right] = v_0^2 - 2gR$$

or $v_0^2 = 8gR - \frac{3NR}{m}$

$\therefore v_0 = \sqrt{8gR - \frac{3N}{m}}$